SURVEY REPORT · QUESTIONNAIRE DESIGN & IMPLEMENTATION

Profiling Data Science Practitioners in Algeria

> *An analysis of skills, tools, and practices of data science professionals and students based on a structured survey*

SAMPLE SIZE	DATA COLLECTION
62 respondents	Structured online questionnaire
DOMAIN	**ACADEMIC YEAR**
Data Science & ML	2025 – 2026

62	74%	80%	71%
RESPONDENTS	STUDENTS	USE PYTHON	MALE

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SECTION I

Introduction

The rapid global growth of data science and machine learning has sparked considerable interest in understanding who the practitioners of these disciplines are, how they learn, what tools they adopt, and how they integrate into the professional landscape. In a developing economy such as Algeria, these questions carry additional weight: the answers shape educational policy, corporate investment, and the country's capacity to participate in the knowledge economy.

This report presents the results of a structured survey conducted among **62 individuals** from the Algerian data science community, including university students, researchers, and working professionals. The instrument comprised **43 variables** covering demographic characteristics, academic background, technical competencies, professional practices, and infrastructure access.

The work follows the methodological framework of applying each of the ten prescribed steps: problem definition, target population specification, sampling plan, questionnaire construction, data collection, coding, quality control, non-response adjustment, statistical analysis, and reporting.

> ***Research question:*** *What is the current profile of data science practitioners in Algeria in terms of demographics, academic formation, technical tools, and professional integration — and what structural gaps constrain the growth of this community?*

METHOD	Non-probability convenience sampling — self-administered online questionnaire
INSTRUMENT	43-variable questionnaire — single-choice, multiple-choice, and dichotomous closed questions
PERIOD	April/May 2026
SAMPLE SIZE	n = 62 valid respondents
ANALYSIS	Univariate descriptive statistics — frequencies, proportions, bar and donut charts

SECTION II

Study Objectives and Constraints

2.1 Study Objectives

The survey was designed to address five interconnected objectives:

#	OBJECTIVE	TARGET VARIABLES
1	Build a socio-demographic profile of practitioners	Age, gender, student status
2	Map the dominant languages and frameworks	Python, R, MATLAB, Scikit-learn, TensorFlow
3	Assess programming and ML experience levels	Years of coding, years of ML usage
4	Measure professional integration of ML methods	Role, sector, company size, employer ML adoption
5	Identify preferred learning and training sources	MOOC platforms, university courses

2.2 Constraints

Every survey operates within constraints that must be anticipated and documented:

BUDGETARY & TEMPORAL

No dedicated funding was available. Online administration minimised distribution and data-entry costs. A limited collection window imposed a convenience sample, with a median response time of approximately 8–12 minutes.

SAMPLING FRAME

No exhaustive register of Algerian data science practitioners currently exists, precluding strict probability sampling. The non-response rate is significant for sensitive variables such as salary and company size — analysed in Section IX.

SECTION III

Target Population and Sampling Plan

3.1 Target Population

The **target population** is defined as all individuals residing in or operating from Algeria who actively practise, study, or research data science, machine learning, or closely related disciplines (applied statistics, programming, and artificial intelligence).

3.2 Sampling Frame

Because no official register of this population exists, the **sampling frame** was constituted from participants who engaged with publicly available data science survey channels. This frame, while accessible and domain-relevant, is not exhaustive.

3.3 Sampling Method and Justification

Given the absence of a complete sampling frame, the method adopted is **non-probability convenience sampling**, complemented by digital snowball diffusion (respondents sharing the questionnaire within their networks). This choice is justified by the exploratory and descriptive nature of the study; strict probabilistic inference to the broader population is therefore not claimed.

TARGET POPULATION	Algerian data science practitioners and students
SAMPLING FRAME	Algerian survey participants
METHOD	Non-probability convenience + digital snowball diffusion
SAMPLE SIZE	n = 62 valid statistical units
STATISTICAL UNIT	The individual respondent
COLLECTION MODE	Self-administered structured online questionnaire

SECTION IV

Questionnaire Design

4.1 Structure

The questionnaire follows the three-part canonical structure prescribed in the methodological literature:

PART	CONTENT	VARIABLES
I — Introductory block	Survey purpose, confidentiality, estimated duration, instructions	1
II — Individual characteristics	Age, gender, student status, education level, professional status	6
III — Survey-object questions	Languages, ML tools and frameworks, experience, cloud platforms, salary, industry sector	36

4.2 Question Types

The instrument mobilises the four closed-question types described in the course:

TYPE A — SINGLE-CHOICE

Ex.: "What is your gender?" — One modality selected from a mutually exclusive list. Used for age bracket, gender, education level, and remuneration.

TYPE B — MULTIPLE-CHOICE

Ex.: "Which programming languages do you use regularly?" — Several simultaneous responses permitted. Applied to tools, platforms, and ML methods.

TYPE C — DICHOTOMOUS

Ex.: "Are you currently a student?" — Yes / No only. Suited to binary variables such as student status, research publication, and GPU usage.

TYPE D — OPEN QUANTITATIVE

Ex.: "How many years have you been writing code?" — Free numerical entry subsequently grouped into ordered brackets for analysis.

4.3 Formulation Principles

Following the rules synthesised in the reference notes (Singly, 2005; Fenneteau, 2002), the questionnaire respected: one idea per question, positive phrasing without double negation, general-to-specific ordering within each thematic block, sensitive items (salary) placed at the end, and an "Other (please specify)" option offered where the list might be incomplete.

4.4 Pre-test

Before full deployment, an informal pre-test was conducted on a small restricted group. It led to the reformulation of two ambiguous items, the adjustment of response scales, and confirmed an average completion time of 8–12 minutes — within the acceptable range for this type of instrument.

SECTION V

Socio-Demographic Characteristics

5.1 Gender and Student Status

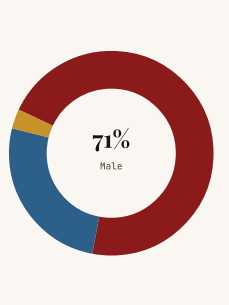


Fig. 1 — Gender (n=62)

- ● Male 44 (71%)
- ● Female 16 (26%)
- ● Not stated 2 (3%)

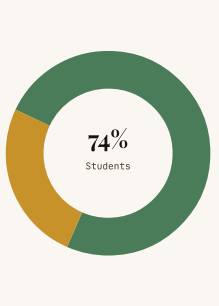


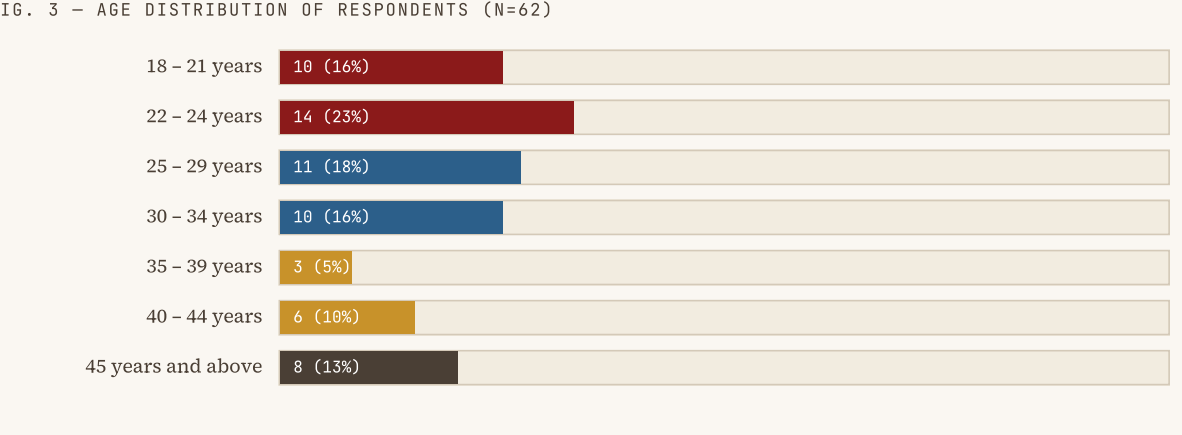
Fig. 2 — Student Status (n=62)

- ● Currently a student 46 (74%)
- ● Not a student 16 (26%)

The sample is predominantly **male** (71%, n=44), with women representing 26% (n=16). This imbalance mirrors the wider gender gap in STEM communities worldwide and signals an area requiring targeted policy action in Algeria. Three quarters of respondents (74%, n=46) are currently enrolled as students at university or graduate level, confirming the predominantly academic nature of the sample.

5.2 Age Distribution

FIG. 3 — AGE DISTRIBUTION OF RESPONDENTS (N=62)



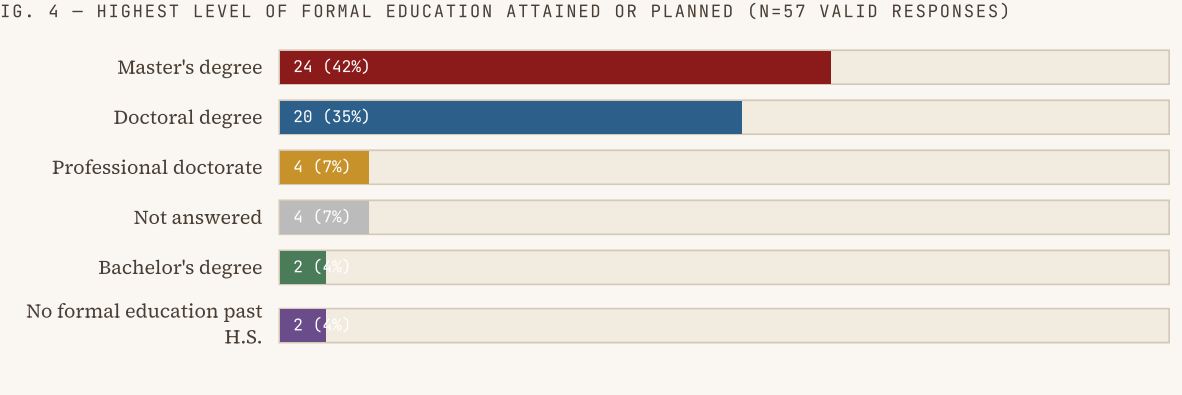
The **22–24** age bracket is the most represented (14 individuals, 22.6%), followed by 25–29 (11 individuals, 17.7%). Overall, **56%** of respondents are under 30 years old, reflecting the youth and dynamism of the Algerian data science community. The presence of respondents aged 40 and above (16%) introduces a degree of professional seniority into the sample.

SECTION VI

Academic Profile and Training Pathways

6.1 Highest Level of Education

FIG. 4 — HIGHEST LEVEL OF FORMAL EDUCATION ATTAINED OR PLANNED (N=57 VALID RESPONSES)

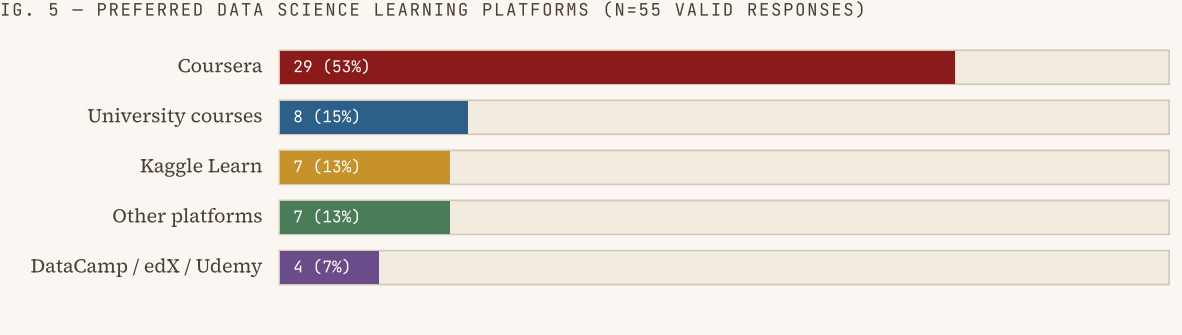


Education levels are **remarkably high**: 77% of respondents hold or plan to attain a Master's or Doctoral degree. Specifically, 24 respondents (42%) are at Master's level and 20 (35%) at Doctoral level. This distribution reveals a community that is strongly anchored in academic research and explains the high publication rate reported below.

> ***Key insight:*** *The co-occurrence of high education levels and significant publication rates (42% of respondents have published academic work) suggests that the Algerian data science community is primarily anchored in academic research rather than industry practice — a pattern further corroborated by the low professional employment rate in Section VIII.*

6.2 Preferred Learning Platforms

FIG. 5 — PREFERRED DATA SCIENCE LEARNING PLATFORMS (N=55 VALID RESPONSES)



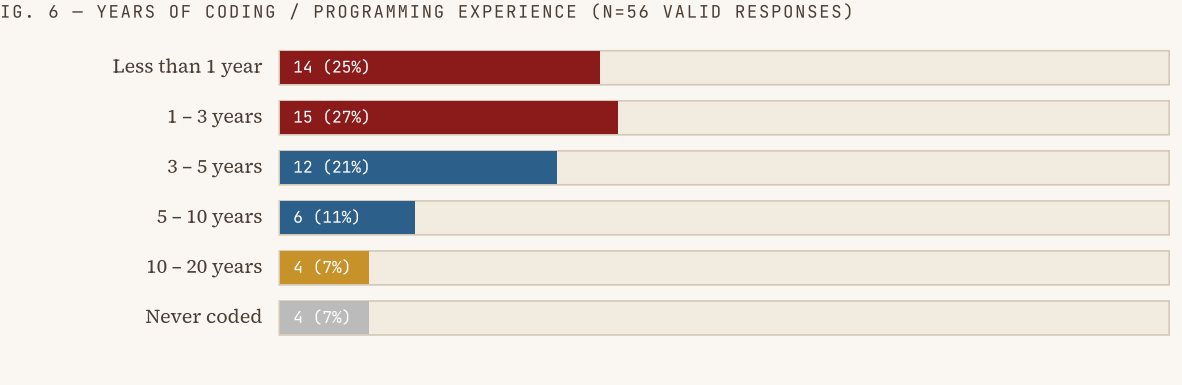
Coursera is the dominant learning platform (53%, n=29), followed by university courses (15%) and Kaggle Learn (13%). This hierarchy highlights the central role of **MOOCs** in bridging the gap between Algerian educational institutions and the rapidly evolving skill requirements of data science. The fact that Coursera outranks formal university courses is itself significant: online platforms have substantially complemented — and in some cases supplanted — traditional academic pathways.

SECTION VII

Technical Competencies and Tools

7.1 Years of Programming Experience

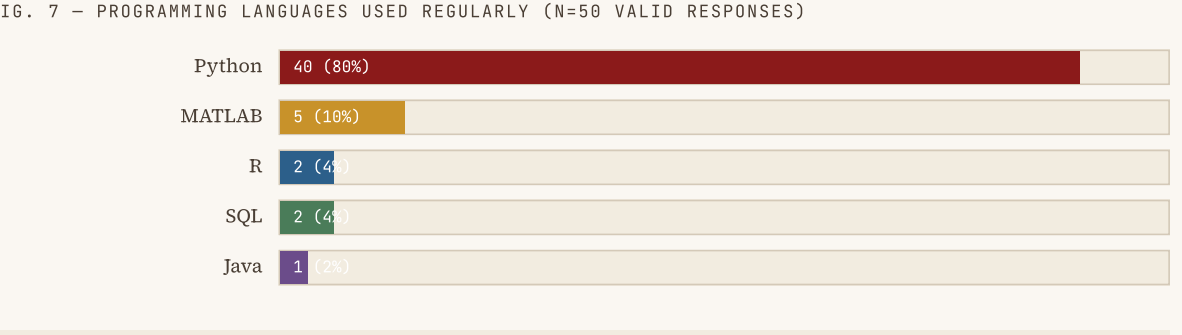
FIG. 6 — YEARS OF CODING / PROGRAMMING EXPERIENCE (N=56 VALID RESPONSES)



Programming experience is largely **recent**: 52% of respondents have fewer than 3 years of experience, and 7% have never written code. This is consistent with the young, predominantly student composition of the sample.

7.2 Programming Languages in Regular Use

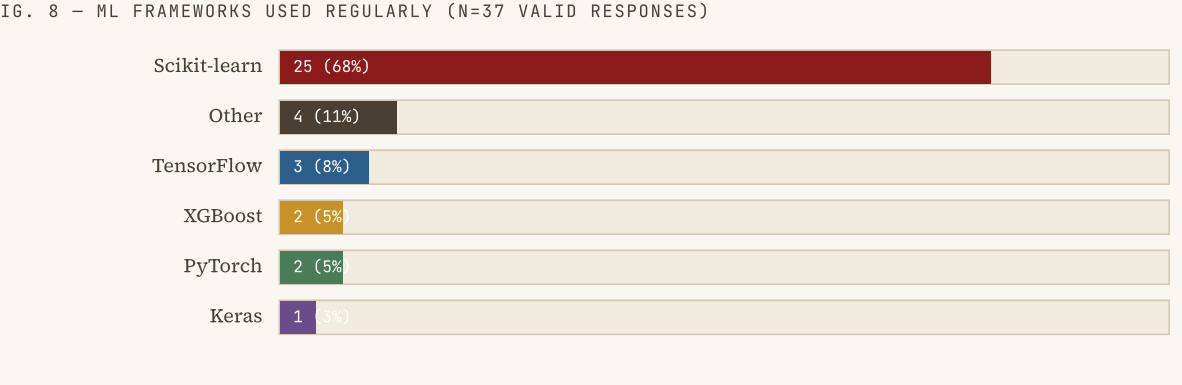
FIG. 7 — PROGRAMMING LANGUAGES USED REGULARLY (N=50 VALID RESPONSES)



> ***Python*** *dominates with* ***80%*** *regular usage, confirming its standing as the universal language of data science worldwide. The notable presence of* ***MATLAB*** *(10%) reflects its continued use in Algerian engineering curricula, where it remains a standard teaching tool. Near-zero SQL usage (4%) is a significant curricular gap, as database skills are essential for most professional data science roles.*

7.3 Machine Learning Frameworks

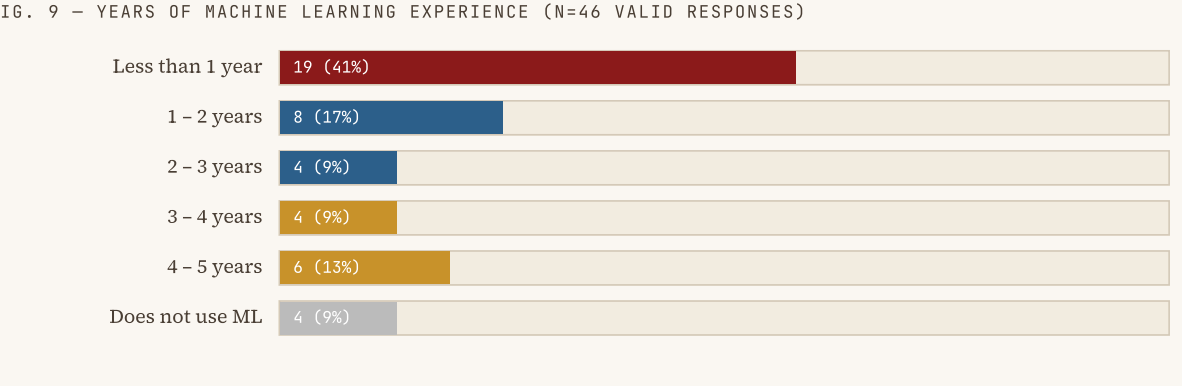
FIG. 8 — ML FRAMEWORKS USED REGULARLY (N=37 VALID RESPONSES)



Scikit-learn leads with 68% (n=25), reflecting its status as the reference library for classical machine learning. TensorFlow, XGBoost, and PyTorch trail significantly — a finding directly linked to the very restricted GPU access documented below.

7.4 Machine Learning Experience

FIG. 9 — YEARS OF MACHINE LEARNING EXPERIENCE (N=46 VALID RESPONSES)



ML experience is recent for the majority: **58%** have fewer than 2 years of ML practice. A growing segment (13%) has 4–5 years of experience, attesting to an emerging layer of more senior practitioners within the community.

7.5 Infrastructure and Development Environment

TOOL / TECHNOLOGY	USED	RATE	OBSERVATION
JupyterLab	Yes	Majority	Reference IDE for data science
Kaggle Notebooks	Yes	Significant	Free cloud; popular with students
Matplotlib	Yes	Majority	Standard visualisation library
GPU (specialised HW)	Yes	3% (2/62)	Very limited — critical infrastructure gap
Amazon Web Services	Yes	~2%	Near-absent cloud usage
MySQL (databases)	Yes	Marginal	Low use of data warehousing tools

A critical structural finding: only **2 respondents out of 62** use GPUs for model training. This directly constrains deep learning practice and largely explains why classical ML (Scikit-learn) so greatly outranks deep learning frameworks (PyTorch, TensorFlow) in adoption.

SECTION VIII

Professional Practices and ML Integration

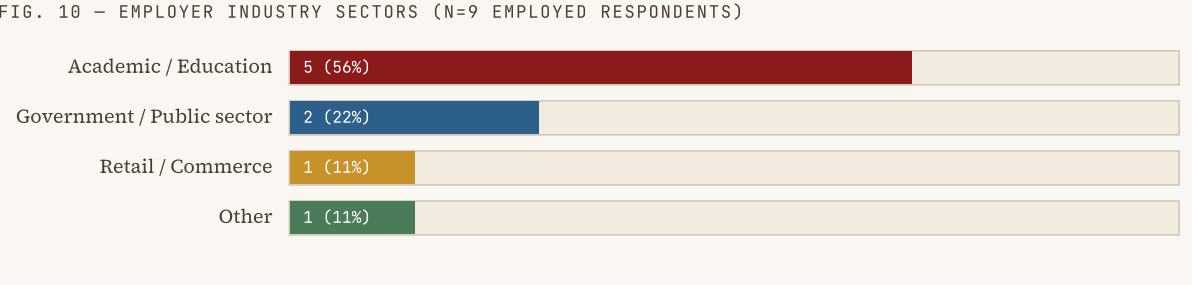
8.1 Professional Roles

82% of respondents (n=51) did not fill in a professional title, structurally consistent with the majority student composition. Among those who declared a role:

PROFESSIONAL TITLE	COUNT	%
Research Scientist	3	4.8%
Teacher / Professor	2	3.2%
Data Analyst	2	3.2%
Currently not employed	2	3.2%
Software Engineer	1	1.6%
Data Scientist	1	1.6%
Not stated (students)	51	82.3%

8.2 Industry Sectors

FIG. 10 — EMPLOYER INDUSTRY SECTORS (N=9 EMPLOYED RESPONDENTS)



8.3 Employer Adoption of Machine Learning

Among employed respondents, **4 explicitly state that their employer does not use machine learning**. Only one employer is actively exploring ML methods, and one has models in production for less than two years. This low adoption rate reveals a significant gap between the technical competencies of the workforce and the actual practices of Algerian companies.

> *This gap between workforce competency and employer adoption is both a challenge and an opportunity: it signals a market where professionally trained data scientists are scarce relative to the community's collective training level, and where early movers could gain significant competitive advantage.*

8.4 Annual Remuneration

The non-response rate on the salary variable is very high: **54 missing values out of 62** (87%). Among the 8 respondents who disclosed their remuneration:

ANNUAL COMPENSATION BRACKET (USD)	COUNT
$0 – $999	4
$1,000 – $1,999	2
$4,000 – $4,999	1
$20,000 – $24,999	1

Declared remunerations are predominantly low, reflecting either students with supplementary income or early-career professionals in an emerging market where data science salaries have not yet reached international levels.

SECTION IX

Data Quality Control and Limitations

9.1 Partial Non-Response Analysis

VARIABLE	MISSING	RATE	PROBABLE CAUSE
Annual remuneration	54 / 62	87%	Sensitive; many students have no income
Company size	53 / 62	85%	Not applicable to students
Professional title	51 / 62	82%	Not applicable to students
ML use in company	53 / 62	85%	Not applicable to students
Education level	5 / 62	8%	Uncertainty about planned pathway
Coding experience	6 / 62	10%	Difficulty accurately self-assessing

9.2 Treatment of Non-Responses

In accordance with the prescribed methodology, partial non-responses were treated by **listwise deletion** for each univariate analysis. Valid sample sizes are reported in all figures. No imputation was performed for professional variables (salary, company size), as the non-response is **structural** — the majority of respondents are students to whom these questions do not apply.

9.3 Validity and Reliability

Internal validity was verified through inter-variable consistency: student profiles declaring limited coding experience are coherent with their education levels and absence of professional income. **External validity** is limited by the non-probability sampling method — results are descriptive and should not be generalised to the full Algerian data science population without a probability-based replication study.

> ***Cronbach's Alpha:*** *The reliability coefficient recommended for Likert-type attitudinal scales (acceptable threshold $\alpha > 0.7$) would be applicable to any future version of this instrument incorporating scaled opinion items. For the factual closed questions used here, internal consistency is confirmed through logical cross-tabulation of related variables.*

9.4 Survey Limitations

① **Selection bias:** Respondents drawn from online data science channels over-represent digitally active profiles and under-represent practitioners outside these circles.

② **Small sample size:** n=62 limits statistical precision and precludes robust sub-group analyses.

③ **High professional non-response:** Employment variables are sparsely answered, constraining the professional-dimension analysis.

④ **Cross-sectional design:** A single collection point prevents longitudinal analysis of skill development over time.

⑤ **Social desirability bias:** Respondents may have over-reported tool usage or experience levels when filling in the questionnaire.

SECTION X

Conclusions and Recommendations

10.1 Summary of Key Findings

#	FINDING	KEY INDICATOR
1	Predominantly young and male community	56% under 30; 71% male
2	Very high academic formation level	77% hold or are pursuing a Master's or PhD
3	Python is the universal tool	80% of programmers use it regularly
4	ML experience is recent and still developing	58% have under 2 years of ML practice
5	Critical infrastructure gap	Only 3% of respondents have GPU access

10.2 Recommendations

R1 **Develop shared computing infrastructure.** University GPU clusters and subsidised cloud access would directly address the hardware gap that currently prevents deep learning adoption, enabling Algerian researchers to compete internationally.

R2 **Strengthen university–industry transfer.** The low ML adoption rate among employers calls for structured internship programmes, industry partnership agreements, and continuing professional development curricula targeting managers and decision-makers.

R3 **Promote gender inclusion.** With only 26% female respondents, targeted initiatives — scholarships, mentoring, and visibility campaigns — are needed to reduce the gender gap in Algerian data science.

R4 **Integrate data engineering into curricula.** Near-zero SQL and cloud storage usage signals a significant curricular gap, given that these skills are standard requirements in professional data science roles.

R5 **Improve future survey design.** A stratified probability sample and a formal questionnaire pre-test would substantially reduce non-response rates and strengthen representativeness in subsequent editions of this survey.

10.3 Methodological Contribution

This report is based on a survey in which data were collected from the Algerian data science community following a full research workflow that includes objective definition, population framing, non-probability sampling, structured questionnaire design, online administration, coding, quality control, non-response treatment, and univariate descriptive analysis, to a real dataset from the Algerian data science community. It concretely illustrates the challenges inherent in social surveys: high non-response on sensitive variables, selection bias from digital distribution, and the need to balance informational depth with respondent burden.

SECTION XI

References

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n = 62 respondents
43 variables analysed
May 2026